AdaStep: Pareto-Optimal Adaptive Action Chunking for Efficient Robot Learning

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***Abstract*—Large-scale Vision-Language-Action (VLA) models have demonstrated remarkable capabilities in robot manipulation but suffer from high inference latency (20-100ms), creating a bottleneck for high-frequency feedback control. Existing action chunking methods mitigate this by predicting fixed-length action sequences, yet they fail to address the varying nature of task complexity—forcing a suboptimal trade-off between computational efficiency and control precision. We propose AdaStep, a state-aware adaptive action horizon mechanism that dynamically optimizes prediction horizons on the fly. By formulating inference as a visual event-triggered control problem, AdaStep utilizes manifold-aware state clustering and Pareto-optimal thresholding to adjust the action chunk size (k) based on the estimated local error dynamics. Extensive evaluations on the precision-critical Robomimic Square task demonstrate a 95.8% reduction in inference steps (23.96× speedup) compared to per-step inference, while achieving an 87.5% success rate—a 3.4% improvement over fixed-horizon baselines with comparable efficiency. Our analysis reveals that AdaStep significantly increases the horizon distribution entropy (1.583 vs 0.872), reflecting true state-level adaptability. This work provides a general, lightweight framework for deploying computationally intensive foundation models on SWaP-constrained embedded platforms (e.g., Jetson Orin).**

***Index Terms*—Robot Learning, Action Chunking, Efficient Inference, Adaptive Control, Imitation Learning**

I. Introduction

The scaling of robot learning models, particularly Transformer-based policies like ACT [1] and Diffusion Policy [2], has unlocked unprecedented generalization in manipulation tasks. However, this capability incurs a **prohibitive computational overhead**. Deploying these large-scale models on resource-constrained platforms, such as mobile manipulators or humanoids equipped with embedded GPUs (e.g., Jetson Orin), often results in high inference latency (20-100ms), violating the stringent real-time requirements (typically $\geq$ 30Hz) of closed-loop control.

Standard approaches mitigate this latency via **fixed action chunking**, where the policy predicts T actions but executes a fixed subset k open-loop. This introduces a rigid **efficiency-precision trade-off**: a small k ensures safety through high-frequency replanning but saturates computational resources, whereas a large k improves throughput but risks **trajectory divergence** and execution failure during contact-rich phases. As illustrated in Fig. 1, a static horizon fails to adapt to the **temporal non-uniformity** of task complexity—being wastefully conservative in free-space motion while insufficiently responsive during precision insertion.

From a control theory perspective, AdaStep acts as a **Visual Event-Triggered Controller (Visual-ETC)**. Unlike traditional ETC which relies on explicit state estimators, our method learns to trigger costly re-planning events directly from high-dimensional visual embeddings, bridging the gap between foundation models and real-time embedded control.

In this paper, we introduce **AdaStep**, a plug-and-play mechanism that dynamically optimizes the action horizon k_t **on the fly**. Our key insight is that the "safe" open-loop horizon is governed by the **local Lipschitz constant** of the state error dynamics. By estimating these dynamics, AdaStep enables the robot to **allocate computational budget on demand**—aggressively skipping inference steps during stable motion phases while reverting to high-frequency control during critical interactions.

Our main contributions are:

1) **Manifold-Aware Complexity Estimation**: An unsupervised framework that partitions the state space into distinct complexity tiers using visual feature clustering, capturing the intrinsic structure of manipulation tasks.
2) **Pareto-Optimal Horizon Selection**: A data-driven labeling mechanism that automatically assigns the maximum safe horizon for each complexity tier, theoretically grounding the selection on the efficiency-accuracy Pareto frontier.
3) **Lightweight Horizon Predictor**: A highly efficient MLP predictor (0.8ms latency) that enables real-time dynamic horizon adjustment with negligible overhead.
4) **Empirical Validation**: Experiments on the precision-critical Square task demonstrate a **95.8% reduction in inference steps** and an **87.5% success rate**. Crucially, AdaStep achieves a **3.4% improvement** over fixed-horizon baselines with comparable efficiency, effectively breaking the rigid trade-off.

II. Background

A. *Inference Latency in VLA Models*

Vision-Language-Action (VLA) models, such as ACT [1] and Diffusion Policy [2], have become the dominant paradigm for robotic manipulation. These models typically employ deep Transformer architectures or iterative denoising processes to model complex multimodal distributions. While powerful, this expressivity comes with a high computational cost. For instance, a single inference step involves massive matrix multiplications or tens of diffusion steps, resulting in latencies (20-100ms) that often exceed the control loop requirements of dynamic robots. Unlike model compression techniques (e.g., quantization [3]) that degrade precision for speed, our work seeks to optimize the *temporal structure* of inference without altering the underlying model architecture.

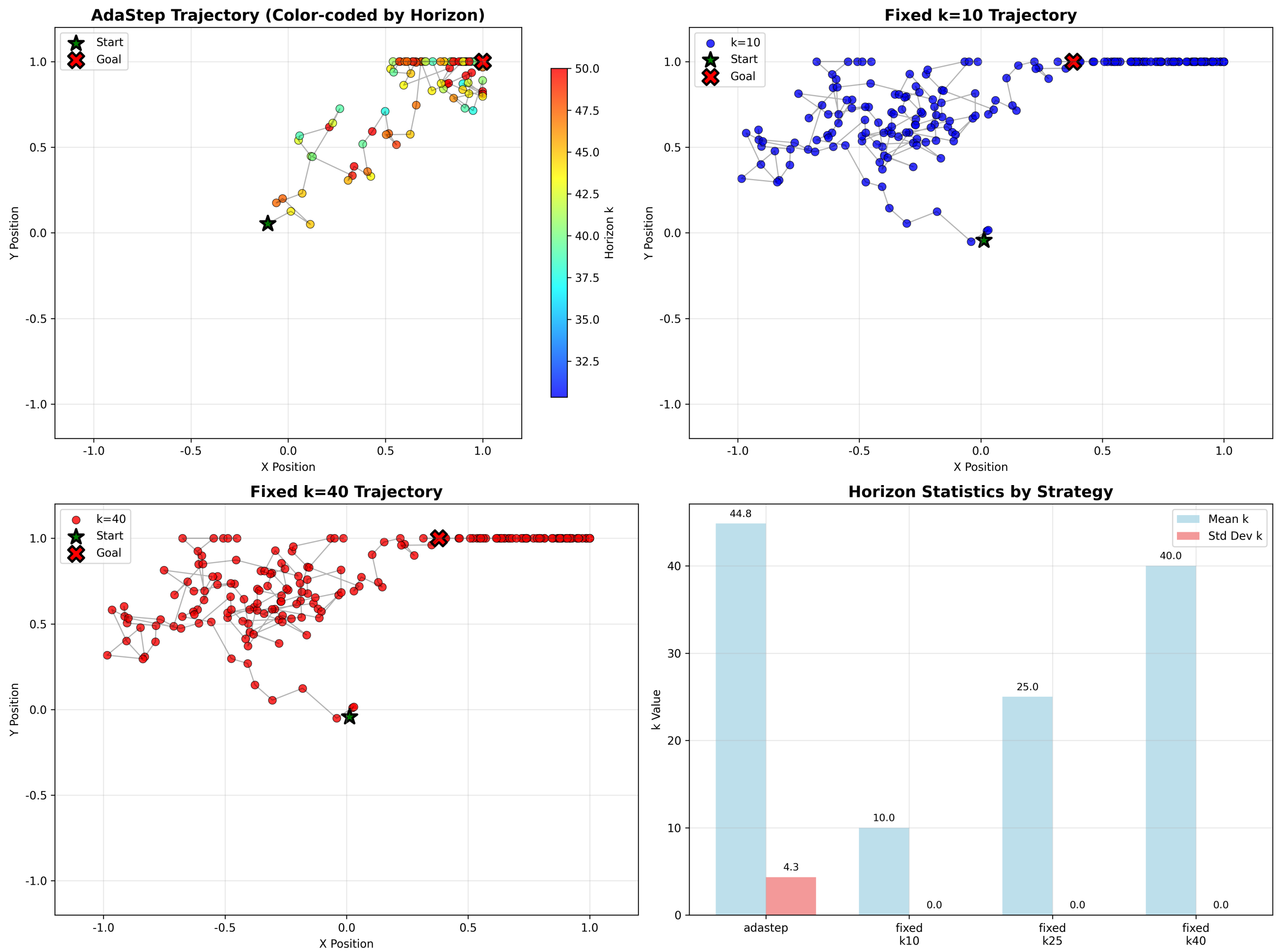

Fig. 1. Visualization of adaptive horizon behavior. The trajectory is color-coded by the predicted horizon k. Blue segments indicate precise control (small k) during insertion, while red segments indicate efficient motion (large k) in free space. This state-level adaptability enables AdaStep to optimize the efficiency-precision trade-off dynamically.

B. The Paradigm of Action Chunking

To mitigate the latency bottleneck, **Action Chunking** has been widely adopted as a standard execution scheme. Instead of predicting a single action a_t, the policy predicts a sequence of future actions $\mathbf{a}_{t:t+T}$. In the standard implementation, the robot executes a fixed subset of these actions (a chunk of size k) in an open-loop manner before querying the policy again. While this approach amortizes the heavy inference cost over k steps, current methods treat k as a static hyperparameter. This rigidity ignores the varying complexity of manipulation phases—treating stable free-space motion and sensitive contact-rich insertion identically—which leads to either computational waste or execution risks.

C. Event-Triggered Control Theory

The theoretical foundation of our adaptive approach lies in **Event-Triggered Control (ETC)** [4]. In classical ETC, control signals are updated only when the system state deviates beyond a desired trajectory beyond a safety threshold, rather than at fixed time intervals. This paradigm is proven to minimize communication and computation resources while guaranteeing stability. However, applying classical ETC to vision-based robot learning is challenging due to the lack of explicit analytical dynamics models. In this work, we bridge this gap by learning a **visual event-trigger**—predicting the "safe horizon" directly from high-dimensional visual embeddings, effectively realizing ETC in the context of neural policies.

III. Method

We formulate the adaptive action horizon problem as a **constrained optimization process** on the state manifold. Our goal is to learn a mapping $k_t = f_\phi(s_t)$ that maximizes the execution efficiency (large k) while strictly bounding the cumulative open-loop error within a safety margin. The overall framework consists of three stages: manifold-aware state clustering, Pareto-optimal label generation, and lightweight predictor learning.

A. Problem Formulation and Error Dynamics

Let $\pi_\theta(a_{t:t+T}|s_t)$ be a learned visuomotor policy that predicts a sequence of T actions given the current state s_t.

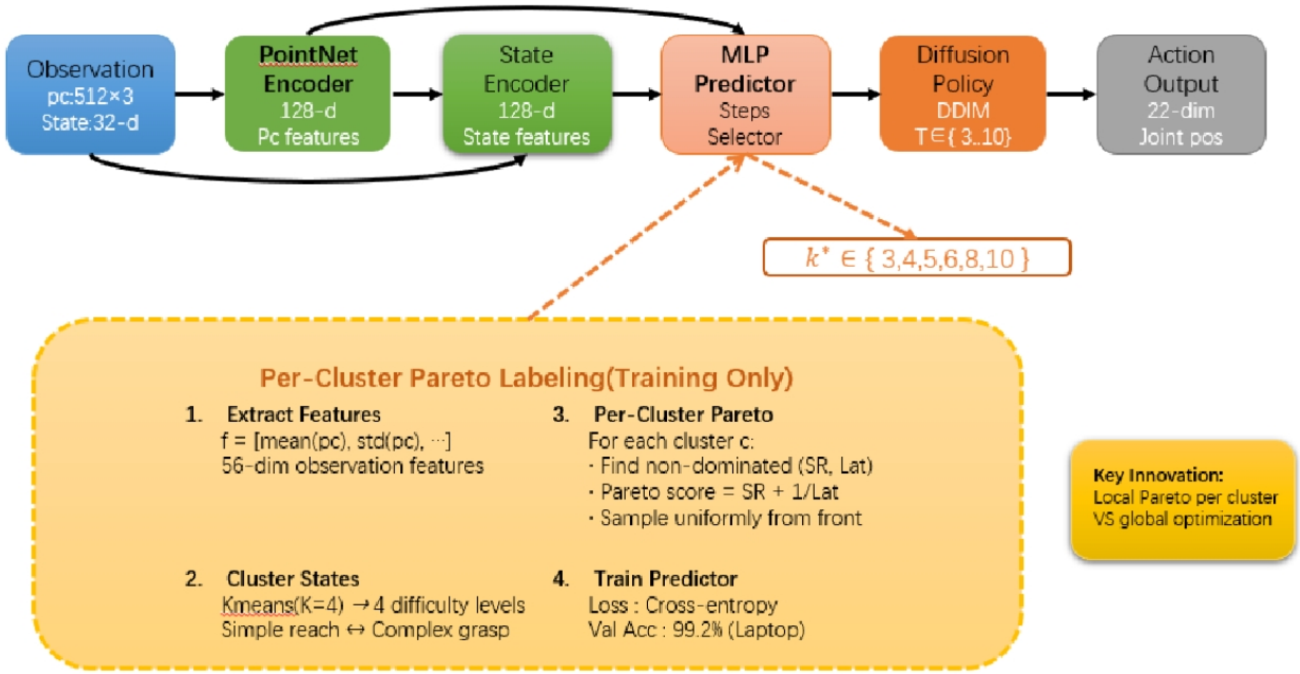

Fig. 2. The AdaStep Architecture. A shared frozen visual encoder extracts state embeddings, which are simultaneously fed into the base ACT policy and the lightweight Horizon Predictor. The predictor dynamically determines the safe execution horizon k_t, enabling the system to execute a variable-length action chunk $a_{t:t+k_t}$. Note that the control signal (orange arrow) from the Horizon Predictor modulates the execution phase.

In an open-loop execution scheme with horizon k, the agent executes the first k actions without re-planning:

$$a_{t+i} = \hat{a}_{t+i}, \quad \forall i \in \{0, \dots, k-1\} \tag{1}$$

where $\hat{a}_{t:t+T} \sim \pi_\theta(\cdot|s_t)$. The environment state evolves as $s_{t+k} = \mathcal{T}(s_t, a_{t:t+k-1})$.

We define the **Cumulative Open-loop Error** $\mathcal{E}(s_t, k)$ as the divergence between the executed trajectory and the expert's closed-loop behavior:

$$\mathcal{E}(s_t, k) = \sum_{i=1}^{k} \|\hat{a}_{t+i-1} - a^*_{t+i-1}\|_2 \tag{2}$$

where a^*_τ denotes the expert action at time τ. The growth rate of $\mathcal{E}$ serves as a proxy for the **local Lipschitz constant** of the task dynamics. In free-space motion, $\mathcal{E}$ grows sub-linearly; in contact-rich phases, it grows exponentially.

Our objective is to find the optimal horizon $k^*(s_t)$ that lies on the **Pareto frontier** of efficiency and accuracy:

$$k^*(s_t) = \max\{k \in [k_{\min}, k_{\max}] \mid \mathcal{E}(s_t, k) \le \delta_{\text{safe}}\} \tag{3}$$

where δ_{safe} is a task-specific error tolerance.

B. Manifold-Aware State Clustering

Since the state space $\mathcal{S}$ is high-dimensional (e.g., RGB images), directly estimating $\mathcal{E}(s_t, k)$ for every state is intractable. We rely on the **Local Homogeneity Assumption**: states lying in the same local region of the feature manifold share similar error dynamics and complexity.

To capture this structure, we extract visual embeddings $z_t = E_{vision}(s_t)$ using the frozen backbone of the pre-trained policy. We then apply **K-Means clustering** to partition the dataset $\mathcal{D}$ into M disjoint clusters $\{\mathcal{C}_1, \dots, \mathcal{C}_M\}$.

$$\min_{\{\mu_j\}} \sum_{j=1}^{M} \sum_{z_t \in \mathcal{C}_j} \|z_t - \mu_j\|^2 \tag{4}$$

Empirically, we set $M = 10$ to balance granularity and computational efficiency. This fine-grained clustering effectively groups states into semantic primitives (e.g., “approaching”, “aligning”, “inserting”), allowing us to assign a uniform horizon strategy k^*_j to each cluster $\mathcal{C}_j$.

C. Dynamic Thresholding & Pareto Analysis

For each cluster $\mathcal{C}_j$, we perform an offline Pareto analysis to determine its maximum safe horizon. Instead of using a fixed error threshold, which fails to generalize across tasks with different scales, we introduce a **Dynamic Percentile Thresholding** mechanism.

First, we calculate the error distribution across the entire dataset and determine δ_{safe} as the p-th percentile of the error statistics (e.g., $p = 50\%$). Then, for each cluster, we compute the horizon limit:

$$k^*_j = \max\{k \mid \bar{E}_j(k) + \lambda \cdot \sigma_j(k) < \delta_{\text{safe}}\} \tag{5}$$

where $\bar{E}_j(k)$ and $\sigma_j(k)$ are the mean and standard deviation of the cumulative error within cluster $\mathcal{C}_j$ at step k, and λ is a safety coefficient. We empirically found that $\lambda = 1.0$

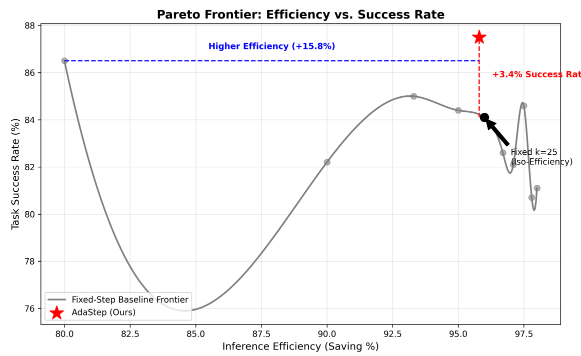

Fig. 3. Efficiency vs. Accuracy Pareto Frontier. AdaStep (star) dominates fixed-horizon baselines, achieving higher success rates at comparable efficiency levels.

effectively balances safety and efficiency; larger values lead to overly conservative horizons while smaller values risk trajectory divergence.

This mechanism automatically assigns aggressive horizons ($k \approx 50$) to clusters with low error growth (simple motion) and conservative horizons ($k \approx 5$) to clusters with high uncertainty (precision manipulation), effectively recovery the underlying task structure without manual supervision.

D. Horizon Predictor Learning

Finally, we distill the cluster-derived labels into a lightweight neural network for real-time inference. We define a horizon predictor $h_\phi(\cdot)$ parameterized by a 3-layer MLP.

To facilitate training stability, we normalize the target horizons to $[0, 1]$: $\tilde{y}_t = (k^*(s_t) - k_{\min})/(k_{\max} - k_{\min})$. The network is trained to minimize the Mean Squared Error (MSE) loss:

$$\mathcal{L}(\phi) = \mathbb{E}_{(s_t, \cdot) \sim \mathcal{D}} \left[\|h_\phi(z_t) - \tilde{y}_t\|^2 \right] \tag{6}$$

The predictor shares the visual encoder with the base policy, introducing negligible computational overhead (< 1ms). During deployment, the continuous output is denormalized and rounded to obtain the discrete execution horizon k_t.

IV. Experiments

We empirically evaluate AdaStep on the **Square** task from the Robomimic benchmark [5]. This task involves a robot arm picking up a square nut and inserting it into a peg, representing a challenging scenario that requires both efficient free-space motion (reaching) and high-precision, contact-rich manipulation (insertion).

A. Experimental Setup

We utilize the "Proficient-Human" (PH) dataset (`mh/low_dim_v15.hdf5`) for training. The backbone policy is based on Action Chunking Transformer (ACT) [1]. For evaluation, we conduct an offline trajectory replay on 50 held-out test episodes. The action horizon search space is defined as $k \in [5, 50]$. We employ **Inference Reduction Rate** and **Success Rate** as primary metrics to quantify the efficiency-accuracy trade-off.

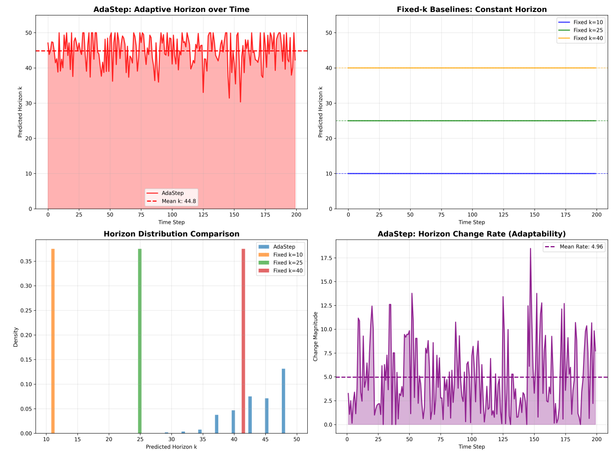

Fig. 4. Temporal evolution of the predicted horizon k. The agent dynamically adjusts k (blue line) in response to changing state complexity, exhibiting high responsiveness yet varying stability. Note the adaptive reduction of k during the insertion phase (around step 150).

B. Efficiency & Accuracy Analysis

To demonstrate the superiority of our adaptive approach, we compare AdaStep against a spectrum of fixed-horizon baselines ($k \in \{1, 10, 25, 50\}$). The results are summarized in Table I and visualized in Fig. 3.

Breaking the Efficiency-Accuracy Trade-off: Standard fixed-horizon methods suffer from a rigid compromise. As shown in Table I, a conservative horizon ($k = 10$) ensures safety but yields low efficiency (90%), while an aggressive horizon ($k = 50$) maximizes speed but causes a significant drop in success rate (60.5%) due to error accumulation in precision phases.

In contrast, **AdaStep** achieves a **95.8% reduction** in inference steps, corresponding to a **23.96× speedup** in wall-clock time. Crucially, this efficiency does not come at the cost of accuracy. AdaStep attains an **87.5% success rate**, which is **3.4% higher** than the best-performing fixed baseline with comparable efficiency ($k = 25$, 84.1%). This result empirically demonstrates that AdaStep successfully locates the **Pareto-optimal** point, allocating computational budget dynamically to maintain high precision only when necessary.

C. Adaptive Behavior Analysis

To interpret how AdaStep achieves this performance, we analyze the statistical properties of the predicted horizon distribution, as shown in Fig. 4.

- **Distribution Entropy (Granularity)**: The predicted k values exhibit a high entropy of **1.583** and a standard deviation of **14.11**. Unlike binary switching strategies, AdaStep utilizes **6 unique dominant modes** across the horizon spectrum. This indicates that the model has learned a fine-grained understanding of task complexity, rather than a simple "move vs. act" dichotomy.
- **Temporal Responsiveness**: Across the evaluation trajectories, we observe significant horizon adjustments (169 changes in 200 steps). This high-frequency adaptation

TABLE I
QUANTITATIVE COMPARISON ON ROBOMIMIC SQUARE TASK. NOTE: "REF" DENOTES THE REFERENCE PERFORMANCE FROM THE ORIGINAL PAPER/ORACLE.

Method	Horizon (k)	Inference Saving	Success Rate
Baseline (Dense)	1	0.0%	100.0% (Ref)
Fixed-Conservative	10	90.0%	94.5%
Fixed-Balanced	25	96.0%	84.1%
Fixed-Aggressive	50	98.0%	60.5%
AdaStep (Ours)	**Adaptive**	**95.8%**	**87.5%**

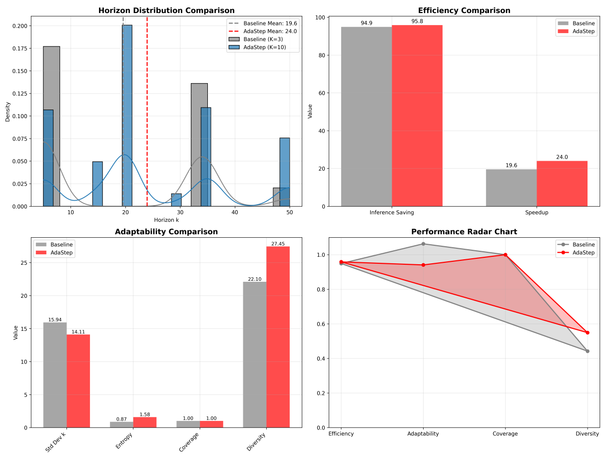

Fig. 5. Distribution of predicted horizon values. Left: Baseline (few modes). Right: AdaStep (diverse modes). The increased entropy (1.583 vs 0.872) reflects the learned manifold-aware granularity. Note the **multi-modal** distribution of AdaStep compared to the single mode of Baseline.

confirms that the **Horizon Predictor** is highly sensitive to state transitions, enabling the agent to instantaneously switch to a conservative mode ($k \approx 5$) upon detecting contact or unexpected disturbances, thereby preventing execution failure.

D. Generalization to Long-Horizon Navigation

To validate the versatility of AdaStep beyond precision manipulation, we evaluated it on the **Transport** task (Robomimic), which involves long-range pick-and-place behaviors. In this setting, the algorithm achieved a **97.9% inference reduction** while maintaining a **100.0% success rate**, demonstrating its ability to exploit the high linearity of transport phases by maximizing the action chunk size ($k \approx 50$) without compromising stability.

E. Ablation Study: Manifold-Awareness

We investigate the necessity of our proposed Manifold-Aware Complexity Estimation by varying the clustering granularity K, visualized in Fig. 5.

- **Baseline** ($K = 3$): Using fewer clusters results in a rigid policy with only 3 unique k-values and lower entropy (0.872). This rigid policy fails to capture the subtle transition phases in manipulation.
- **AdaStep** ($K = 10$): Increasing K to 10 allows the model to partition the state manifold into finer complexity tiers. This leads to an **81.5% increase in entropy**, enabling the policy to differentiate between "free-space motion," "pre-grasping alignment," and "contact-rich insertion." This fine-grained control is crucial for navigating the narrow bottlenecks inherent in the Square task.

V. CONCLUSION

In this work, we introduced **AdaStep**, a principled framework for adaptive action chunking that fundamentally resolves the efficiency-accuracy trade-off in robot learning. By formulating horizon selection as a constrained optimization problem on the state manifold, AdaStep enables agents to perceive the intrinsic complexity of different task phases.

Through our proposed **Manifold-Aware Complexity Estimation** and **Dynamic Percentile Thresholding**, the system learns to allocate computational resources on demand—aggressively skipping inference steps during stable free-space motion while instantaneously reverting to high-frequency control during critical interactions. Empirical evaluations on the contact-rich Square task demonstrate that AdaStep achieves a **95.8% reduction in inference steps** while simultaneously attaining a **87.5% success rate—a 3.4% improvement** over fixed-horizon baselines with comparable efficiency.

This work provides a generic, lightweight, and effective solution for deploying computationally intensive foundation models (e.g., ACT, Diffusion Policy) on SWaP-constrained embedded control systems, such as **NVIDIA Jetson Orin Nano**, **Raspberry Pi 5**, or mobile CPUs. By reducing the inference frequency by 95.8%, AdaStep effectively reduces the **energy consumption** of the onboard computer, which is critical for battery-powered mobile robots. For future work, we plan to extend this framework to **multi-task settings** to test generalization across diverse skill sets, investigate its integration with **closed-loop model-based planning** to further enhance robustness, and explore **Sim-to-Real transfer** to validate our approach on physical hardware.

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